

Unveiling the Eclipse Effect:

Impacts on Pollination Routes of Western Honey Bee Colonies

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In the animal kingdom, there are multiple different types of animals. Each of them are suited for different habitats, environments, and ways of living. Honeybees are unique from most animals as they stay in a consistent pattern of movement and routes in their pursuit to collect nectar for the creation of a substance called honey. This begs concerns for unorthodox factors such as a solar eclipse, that can “block the sun for a period of time, leaving a darkness upon the habitats of those honeybees” (Sinu). Scientists such as zoologists, biologists, and behaviorists have been researching this topic for a while. This can help bring new light to animal behaviors and how animals think along with research on space and how solar eclipse can have an effect on living beings. This research could help scientists understand animal behavior to learn things about how animals interact and learn which would help broaden our understanding of the world around us. It would also prevent any dangers that would come from animals being scared of having anxiety over the eclipse.

The western honey bee (*Apis mellifera*) is a pollinator in various ecosystems and is important for agricultural pollination. Scientists study honey bee behavior during eclipses to understand how sudden changes in light and temperature affect their pollination routes and activity levels. Solar eclipses can cause bees to alter their foraging and pollination patterns. Observations indicate that bees may reduce activity, or even return to their hives (Sinu). Bees rely on the sun for navigation and time-based foraging cues. The eclipse may interfere with their internal clock and navigation mechanisms. These behavioral changes occur specifically during solar eclipses, with the effects more pronounced during total or near-total eclipses when light levels drop significantly. The effects are observed in ecosystems that rely on honey bees for

pollination “such as meadows and forests” (Adara). Since bees may stop foraging during an eclipse, this could temporarily affect pollination, especially if the eclipse occurs during pollination.

Through time and dedication, Scientists have gotten closer to their goal of learning the impact solar eclipses have on pollination routes of western honey bee (*Apis Mellifera*) colonies. This is dealing with animal behavior such as bees and how they act and how they react to a new environment and factors such as a solar eclipse. When dealing with astronomy, it comes with how the solar eclipse works and how it would affect the environment. As there are multiple phases of a solar eclipse as a total eclipse would be different than a partial eclipse, there needs to be data collecting on what specific time and eclipse phase would affect the bees and their routes. Scientists play a critical part in analyzing and collecting data for these experiments and fields. Without them, the knowledge gained from collecting and analyzing these fields. For this type of research, zoologists, biologists, and behaviorists. They collect their data through observation, GPS tracking, and “video to keep track of the bees and ensure that they can get the correct data” (Hartstone). They track their patterns, how many bees have been in a specific flower, and how they interact with their environment before and during the eclipse.

The goal of the research is to answer What impact do solar eclipses have on pollination routes of western honey bee colonies? It would benefit humankind to understand more about animal behavior and learn from them. Bee activity impacts habitats and other ecosystems that rely on plants to thrive and removing those pollination routes could harm them. This study would look into the bees behavior regarding the solar eclipse using GPS trackers to monitor the bees “location and routes throughout a closed known habitat” (Ritson, Ranglack and Bickford). Using

observation tactics and monitoring, it would allow for easy data collection on known patterns and routes that would offer an idea on bees navigation.

Literature Review

Impact on Animal Psychology

When a solar eclipse occurs, it causes changes in light which causes animals to respond differently. Branham and Faust answered what occurred during the eclipse and to discuss how this novel event relates to current knowledge of courtship flashing initiation in fireflies? They observed flashing at sites experiencing the totality portion of the eclipse which formed a band 113 km wide. There were observers at “10 sites from Missouri to North Carolina reported observing this flash activity across a path stretching 912 km across the states where remnant late season populations of *P. pyralis* were active” (Branham). They found that possible mechanisms for triggering firefly flashing behavior are discussed in the context of a total solar eclipse. flash activity across a path stretching 912 km across the states where remnant late season populations of *P. pyralis* were active. males of the dusk active firefly species “*Photinus pyralis* (L.) were observed flashing at sites experiencing the totality portion of the eclipse which formed a band 113 km wide” (Branham). The increased flashing indicates the fireflies' behavior suddenly changed when the solar eclipse occurred. Their light, which is only used for night, gives them ideas of possibly mating and moves into the direction of prioritizing that activity when at night. It could relate to bees as their routes could be altered immediately and prioritize that route when turned to darkness.

Ritson, Ranglack, and Bickford discussed how solar eclipses affect animal behavior and assess social media data showing behavioral changes in animal species? The methods they used were analyzing “685 social media observations and 26 published studies to compare animal

behaviors during the 2017 eclipse” (Ritson, Ranglack and Bickford). They found that the most common behaviors noted included increases in vocalization among invertebrates and increases in activity among birds and mammals. Social media showed Invertebrates and birds reduced activity in noise and books/reports show birds decreased in activity but increased in noise. “685 observations of approximately 48 different animal species responding to the 2017 Great American Solar Eclipse, revealing that invertebrates (40%) and birds (35%) were the most frequently reported animals on social media” (Ritson, Ranglack and Bickford). This would bring lots of contradictions to how bees activity has been documented in the past. Vocalization was a consistent activity in the invertebrates and birds which would indicate fear and communication. Bees would have similar responses in using noise to communicate or might disrupt their routes.

Both found significant changes in animals and how they interact with the solar eclipse. Ritson found that birds communicate and cause lots of noise to signal confusion and fear. Fireflies have shown to increase flashing for mating and signaling shown by Branham. These findings show how light disruptions can cause these animals to interact with their environment differently. It emphasizes fear and uncertainty regarding the situation and relates to bee behavior as it overall describes animal behavior and psychology as a whole.

Behavioral Adaptation

Adara and DeNiro investigate how birds navigate and adapt to unique environmental conditions created by solar eclipses? Their methods consist of documenting and analyzing avian activity using bird banding data. Observation, quantitative analyses, including species composition and capture trends are also ways of how they have collected data. From their data, they found that unexpected, record-breaking captures on the eclipse day, which also surpassed the “average trend in captures over time for 18 other banding days in mid-October” (Adara),

highlights the complexity of bird behavior during celestial events. Birds saw confusion and unexpected movement which caused shifts in their patterns. Drawing from this data, birds have shown to experience a dependence for light when flying and when that light has vanished, they panic and move around.

Hartstone-Rose tried to answer how a total solar eclipse impacts the behavior of animals? They observed the animals in the hours preceding, during, and immediately following the eclipse. Live video and audio capture was also used throughout observations to capture behavior during “vocalizations for subsequent quantitative analysis” (Hartstone). Behavioral responses were classified into one or more of four overarching behavioral categories which included, normal, anxiety, novel, and evening. They found “Baseline Behaviors Three of the mammal species (grizzly bears, seals and sea lions) and one of the bird species (kookaburras) did not have noticeable behavioral changes leading up to or during the eclipse” (Hartstone). Approximately three-quarters of the focal species “(13/17, 76%) observed during the eclipse exhibited behaviors that qualitatively differed from their behaviors before or after the eclipse” (Hartstone). The removal of sunlight caused animal activity to change from their typical morning behavior to evening activities.

Hartstone-Rose showed how animals used light to determine the time and once that light vanishes at an odd time such as during a solar eclipse, it messes up their internal body clock. Bees would relate to this as their internal body would change and think of their night routes and paths for pollination or back to the hive. They could rely on the environment to change their patterns to the situation they are facing. Adara and DeNiro showed how birds rely on light for navigation. When light is absent, their navigation and flight are gone. Both Hartstone-Rose, Adara and DeNiro show how animals are dependent on light and they have a hard time adapting

to that light. If those animals have a hard time adapting, bees would as well regarding pollination routes being hard to follow and homing towards the hive been difficult as well.

Bee activity

Sinu, Jose, and Varma discussed what effect does a solar eclipse have on the visitation characteristics of bees to specific plant species, and which phases of the eclipse most critically impact their foraging behavior? It was observed in two sites in India for bee visitation rates during eclipse and non-eclipse days. Bee visits were recorded in person or via video, with data analyzed. They ran t-tests, and mixed-effect models. “Statistical analysis in R software examined the impact of eclipse and non-eclipse days, plant species, and site location on bee visitation behavior”(Sinu). They found that it received visits from “19 bee species, including carpenter bees, honey bees, a stingless bee, and various solitary bees” (Sinu). On the eclipse day, only 11 bee species visited, and during the peak of the eclipse, visits were recorded only on specific species such as “*C. rutidosperma*, primarily from *Tetragonula iridipennis*, *Braunsapis* sp., and *Amegilla zonata*” (Sinu). This proves how the bees' activities reduce as changes from light to darkness impacts what to do next. Bees would stay in fear or confusion and result in night routes back to the hive to respond to the change.

Waiker talked about how a total solar eclipse impacts the foraging behavior and homing abilities of honey bees? Two experiments were conducted to examine the impact of a total solar eclipse on honey bee foraging and homing behavior. In Experiment 1, researchers watch bees , “using alternating observations and manual counters to record bee departures and returns” (Waiker). In Experiment 2, bees marked with fluorescent powder and observed how they got back to hive. During the solar eclipse, honey bee foraging activity decreased during and after

totality compared to the control day, with activity reduced most sharply during total darkness and gradually recovering afterwards. The speed of the bees regarding homing onto specific targets was “slowest during the eclipse compared to morning and evening”(Waiker). Bees leaving pollination routes in favor of the safety of the hive is a common animal behavior that other animals do when a solar eclipse occurs. Reducing activity and not pollinating is a trait that was shown in the experiment. The use of tracking methods really emphasized on how bees moved throughout puts in comparison of the normal routes to the solar eclipse routes.

Both studies correlate with each other as with routes and patterns, bees have a hard time adapting to those changes in light as they depend on that light to help guide them through pollination and finding flowers. Because of how hard it is to find those flowers and routes, bees prioritize getting to the hive and safety before any harm can be done to the routes or the hive. These survival methods are essential to showing why bee activity has slowed down and relates to the studies shown above regarding bees and other animals. All animals have those survival instincts. Thus, they activate when in situations that change drastically.

Methodology

This study would look into the bees behavior regarding the solar eclipse using GPS trackers to monitor the bees “location and routes throughout a closed known habitat” (Ritson, Ranglack and Bickford). Using observation tactics and monitoring, it would allow for easy data collection on known patterns and routes that would offer an idea on bees navigation. Because of the late time for a solar eclipse, observers must be well trained to keep an eye on the bees efficiently. Light and temperature sensors will be in place to track the movements of the bees and other factors such as the specific phase the solar eclipse is in and heat can play a role in bee routes. Cameras will be in place near flowers, trees, and cover the whole area for maximum

coverage. This ensures that nothing is missed by the observers and all data will be recorded for further analysis.

Two months before the solar eclipse starts, observation begins as the bees will be tagged with GPS and the cameras and sensors will be in place. The two months are used to get a sense of the bees' normal routes as they navigate through flowers and other wild plants along with interactions from the environment. The environment will be outside in a closed off area with flowers, the hive, and areas where the majority of bees are around. The sensors will be there to see whether factors such as light and heat also have an impact on the bees. Once the solar eclipse happens, observers will be at their peak and all equipment must be functional. Observers will see their flight patterns and their behavioral shifts that occur for the whole solar eclipse.

Once all the data is gathered, it will be analyzed through looking at correlations, changes in patterns, and other research to draw conclusions. Findings will be presented graphically and visually to show how the bees move and what specially happened. Bee patterns will be shown with a visualization of the area. Graphs will show the sensor data along with the data of bee pollination rates. The video will also show viewers the actual patterns and actions of the bees within the area. There will be lots of time and people that will need to be involved such as transportation to the closed area , observers, data Analysts, and equipment.

Implications

This research could help scientists understand animal behavior to learn things about how animals interact and learn which would help broaden our understanding of the world around us. It would also prevent any dangers that would come from animals being scared of having anxiety over the eclipse. This research explores bee behavior during environmental changes, such as the different phases of an eclipse, which can disrupt their navigation and foraging. Pollinators such

as the western honey bees are crucial to the environment and with understanding their behavior especially during an unusual time such as a solar eclipse. Scientists will understand more about their behavior and potential damage that they could cause to other animals in the environment along with delayed processes as pollination is an important action that helps plants thrive in an environment.

This research provides a unique look upon animals and their behavior in regards to different situations and their interactions with other wildlife. The findings could show ways to preserve pollination routes for bees along with keeping the exercise intact for further bee populations. As such, the benefits this research provides to pollinators and the environment is important and is the reason pollinators such as the western honey bee are able to contribute to the ecosystem.

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Timeline and Budget

Materials Needed

- Transportation costs \$10,000 total
- Observers: One observer for each bee
- Data Analysts
- Research Supervisors
- Printing and reporting costs: \$500.
- Travel for research \$5,000.
- Binoculars:\$500–\$1,000.
- GPS Devices: \$300 per device.
- Rangefinders: \$600.
- Laptops or tablets: \$1,000 each
- Software licenses for ArcGIS: \$1,500 annually.
- Data sheets
- Light and Temperature sensors \$300

Total Material Estimate:
\$104,000–\$110,000.

Timeline of Activities

1. Preparation Phase (February–May 2044)
 - Get equipment, plan travel, closed environment, and train researchers.
2. Data Collection Phase (June -August 2044)
 - Observers will get to the closed environment and start getting comfortable with the bees and start collecting data
3. Data Analysis Phase (September 2044–March 2045)
 - They will clean the data and start analyzing for patterns or any factors that might impact the results
4. Drafting Results (April–June 2045)
 - Getting ready for publication by making charts, graphs, papers, etc.

Total Time : February 2044 to June 2045.